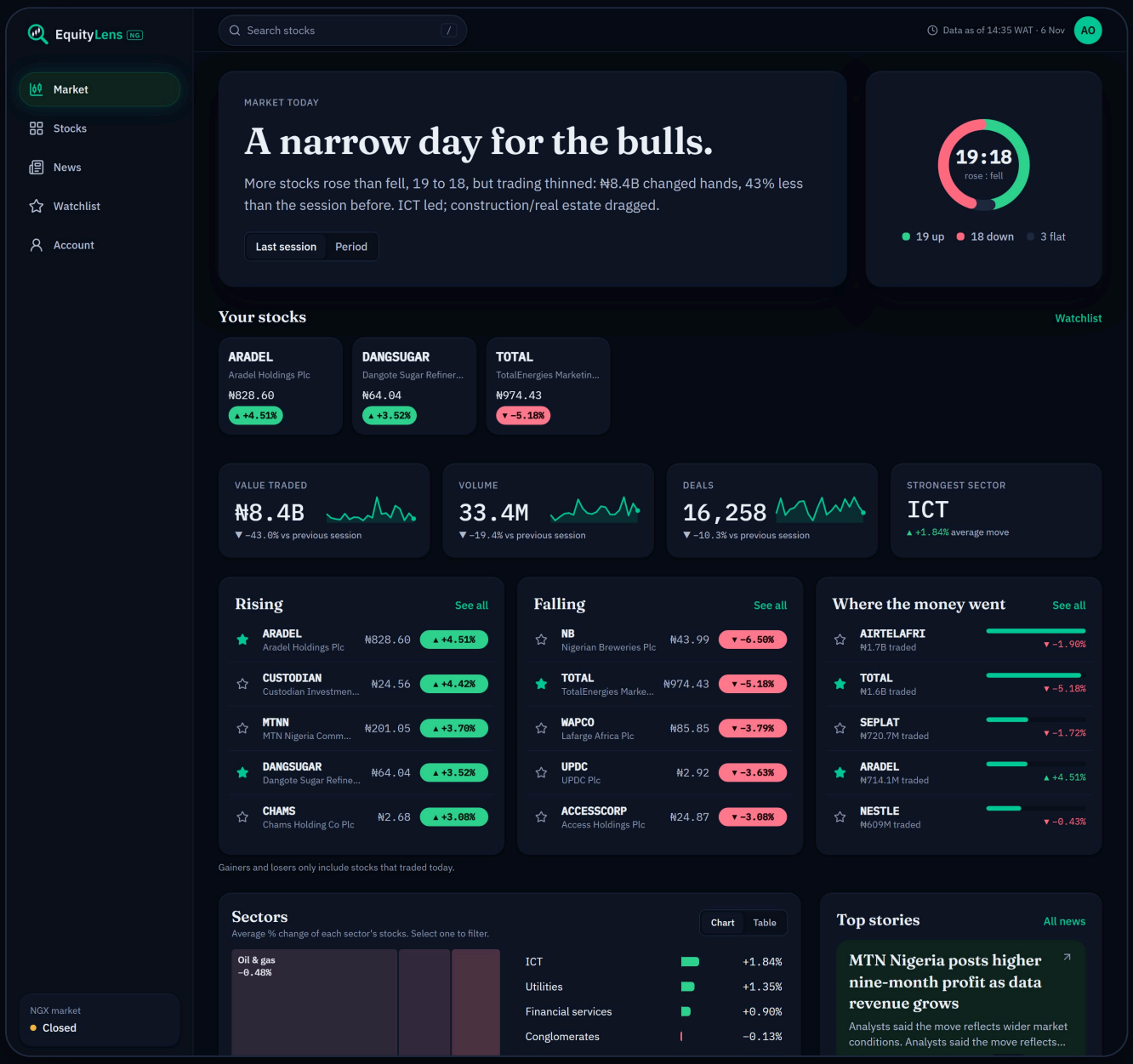
EquityLens NG

DESIGN SYSTEM · VERSION 0.2 · GUIDE UPDATED 6 NOVEMBER 2026

# Lens

## The visual language of EquityLensNG

Calm surfaces, honest data, and colour reserved for meaning. This guide explains the principles, tokens, components and rules behind every EquityLensNG screen.




For designers, developers and stakeholders new to the project

All screens show synthetic test data, not real prices

# What Lens is

Lens is the design system for **EquityLensNG**, a web app that turns Nigerian Exchange (NGX) market data into charts, signals and plain-language explanations. It is the shared set of decisions that makes every screen look, read and behave the same way, in dark mode and in light.

## Three layers

1 **Tokens**

Named values for colour, type, spacing, radius, shadow and motion, kept in one file, `design/tokens.css`. A lint rule blocks raw colour codes.

2 **Components**

Shared building blocks such as buttons and price changes, plus larger pieces built for one screen, such as the chart and the AI card.

3 **Patterns and rules**

How the pieces behave together: loading, data freshness, filtering, errors, keyboard use, accessibility and the voice of the copy.

## Who it is designed for

**Ada, the curious first-timer.** Owns two or three stocks, mostly on mobile, and needs plain language.

**Tunde, the active trader.** Checks the market daily. Wants fast charts, indicators, signals and shareable links.

**Ngozi, the analyst.** Researches sectors and peers for clients. Needs comparisons and exportable data.

All three use the same screens. Depth is revealed on demand, not split into “simple” and “pro” modes.

## Decisions at a glance

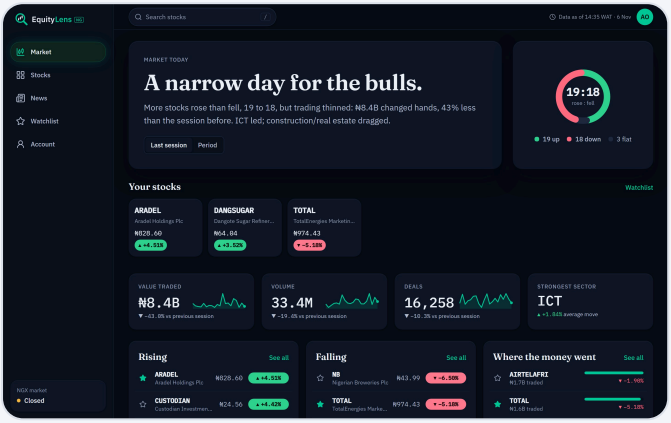
| | |
|---|---|
| Version | 0.2, the “Depth + Editorial headline” direction |
| Palette | Lens Green: #00C896 dark, #00805F light |
| Typefaces | IBM Plex Sans, IBM Plex Mono, Fraunces |
| Themes | Dark by default. Light when chosen, or when the device prefers it |
| Standard | WCAG 2.2 level AA |
| Built with | React, Radix primitives, Tailwind CSS v4, Lightweight Charts, d3-hierarchy, Lucide icons |

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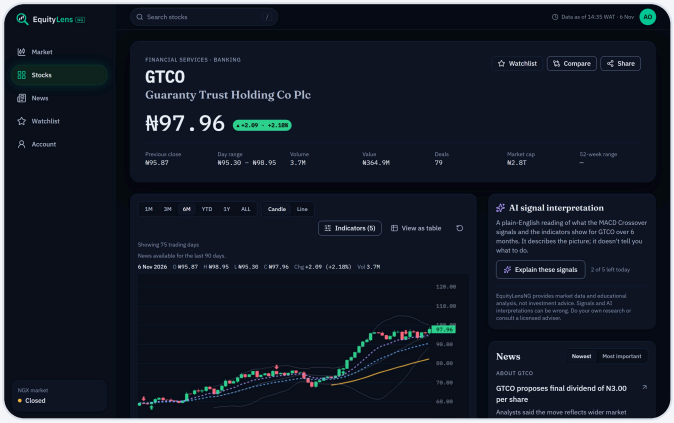
# The product at a glance

Four working screens carry most of the product. Everything in this guide exists to keep them consistent with each other, and with the rest of the app.



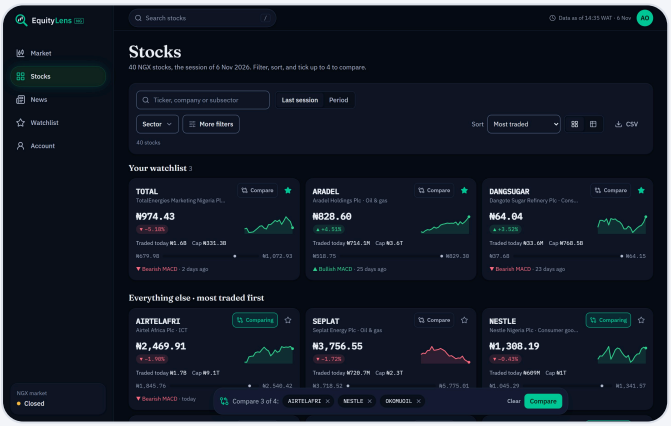

## Market

The day on one page: a plain-language headline, breadth, movers, sectors and top stories.




## Stock

One company: price, chart, signals, peers, news and an AI reading on request.




## Stocks

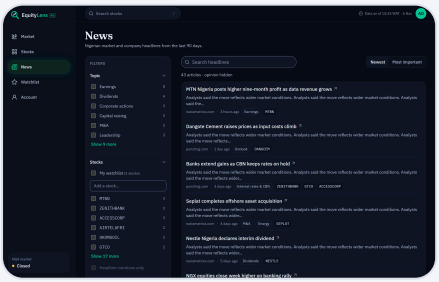
Every listed stock as digest cards or a table, with filters and a tray for starting a comparison.




## Compare

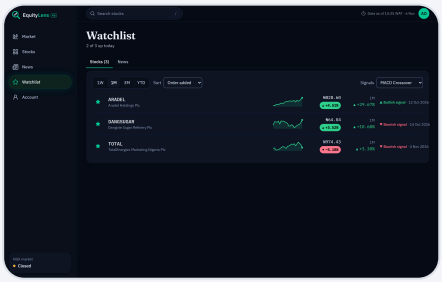
Up to four stocks from a common start, with sector and all-stock averages, side by side.

## Also in the app



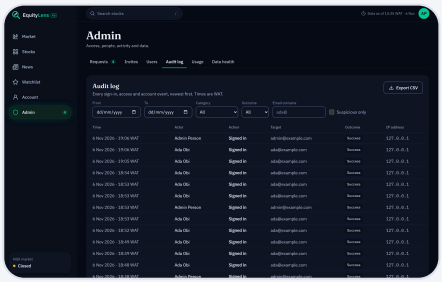

### News

A filterable feed, with counts on every filter.




### Watchlist

Your stocks, their performance and latest signals.




### Admin console

Requests, invites, users, audit log, usage and data health.

Account, Help with a 42-term glossary, and a first-run “Which stocks do you follow?” screen complete the app. Every screen in this guide shows synthetic test data: the company names are real, the prices are random.

# Seven principles

Every rule later in this guide traces back to one of these. When a case is not covered, decide it by asking which principle applies.

01

## Clarity before density

Lead with the one number that matters (price, change, freshness), then reveal depth on demand. Ada and Tunde use the same screens: progressive disclosure, not separate modes.

₦97.96 ▲ +2.08 · +2.18%

02

## Honest data

Always show *when* the data is from, in West Africa Time (WAT). Never imply real time. Show empty, partial and stale states plainly.

Data as of 14:35 WAT · 26 Oct

03

## Direction is never colour alone

Every gain or loss pairs colour with a sign (+ or –) **and** an arrow (▲ or ▼). About 1 in 12 men has a red–green colour vision deficiency.

▲ +2.31% ▼ −0.45% ■ 0.00%

04

## AI is labelled

Anything model-generated sits in the violet AI treatment, with a disclaimer, the model name and a timestamp.

Confidence: medium

Claude Haiku 5.5 · Generated 09:12 WAT

05

## Calm by default, with depth

Layered navy surfaces are lifted by soft elevation rather than outlines. Colour is reserved for meaning: brand action, gain and loss, AI, alerts. Glow belongs to the navigation alone. No gradients or glows on data.

06

## Mobile is first-class

Design at 360 px wide first, then enhance for larger screens. Touch targets are at least 44 px.

07

## Tell the day like an editor

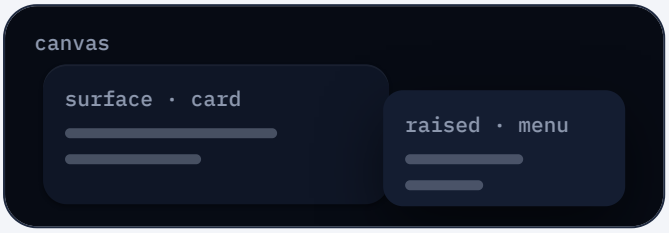
A page leads with a plain-language headline in the display serif and a one-sentence standfirst built from the numbers by a template (never by AI), then the detail.

**A narrow day for the bulls.**

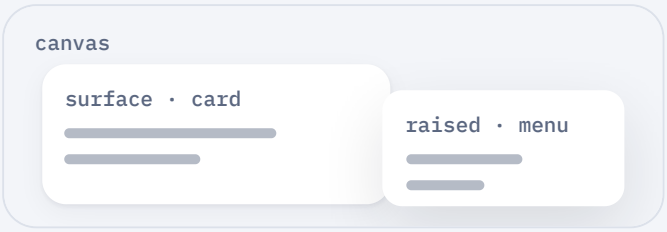
More stocks rose than fell, 19 to 18, but trading thinned: ₦8.4B changed hands, 43% less than the session before.

# Surfaces and text

Dark is the default theme. Light applies when the user chooses it, or when their device prefers light and no choice is saved. A choice saved to an account follows the user to other devices. Contrast was checked against WCAG 2.2: text at least 4.5:1, UI boundaries at least 3:1.




Dark: each step up is a lighter navy plus a soft shadow.




Light: a cool grey canvas with white surfaces lifted by shadow.

| TOKEN AND USE | DARK | LIGHT |
|---|---|---|
| **bg-canvas**<br>Page background | #070B14 | #F3F5F9 |
| **bg-surface**<br>Cards, panels | #0F1626 | #FFFFFF |
| **bg-nav**<br>Sidebar, bottom nav | #0A101C | #FFFFFF |
| **bg-raised**<br>Menus, popovers, dialogs | #141D31 | #FFFFFF + shadow |
| **bg-sunken**<br>Chart plot area, inset wells, bar tracks | #0A101C | #EEF2F7 |
| **bg-hover**<br>Row and item hover | #18223A | #EEF2F7 |
| **bg-selected**<br>Selected row, active nav item, lead story | #10241F | #EBF8F3 |
| **border-subtle**<br>Dividers | #172033 | #E8ECF2 |
| **border-default**<br>Outlines (cards use elevation instead) | #1F2A40 | #DCE1EA |
| **border-input**<br>Form-field boundaries | #61708F 3.5:1 | #848C9C 3.4:1 |
| **text-primary**<br>Body, headings, numbers | Aa #EDF1F8 17.5:1 | Aa #0E1526 18.6:1 |
| **text-secondary**<br>Supporting text, labels | Aa #B4BED0 9.6:1 | Aa #3A4660 9.6:1 |
| **text-muted**<br>Captions, timestamps. **The lowest allowed text colour** | Aa #8D98AE 6.2:1 | Aa #5B6780 5.9:1 |

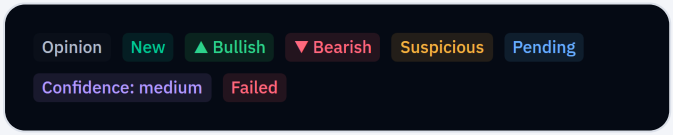
Ratios are the measured contrast for each text and input colour.

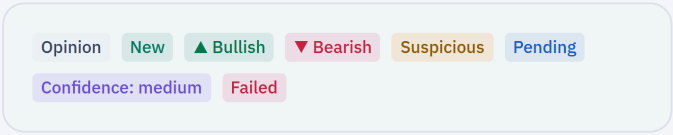
# Brand, market and feedback

Colour carries meaning, so each of these has one job. Gain and loss always travel with a sign and an arrow, so the meaning survives for colour-blind readers and on a black-and-white printout.

| TOKEN AND MEANING | DARK | LIGHT |
|---|---|---|
| **brand**<br>Lens Green: primary action, focus ring, links, active nav icon | #00C896 | #00805F |
| **gain**<br>Price up, always with + and ▲ | #2FD38F | #047A4E |
| **loss**<br>Price down, always with – and ▼ | #FF6B81 | #C42343 |
| **unchanged**<br>No change, shown as 0.00% ■ | #B4BED0 | #3A4660 |
| **danger**<br>Errors, destructive actions | #FF6B81 | #C42343 |
| **warning**<br>Stale data, suspicious activity | #F5B841 | #8F5B00 |
| **info**<br>Neutral notices | #6CB2FF | #1D63C9 |
| **ai**<br>Anything AI-generated | #B49CFF | #6D4ED8 |

## Tints for badges

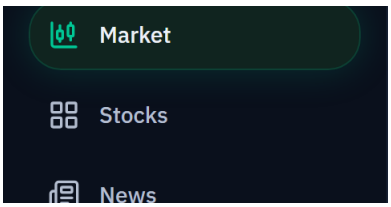



Each colour has a `-subtle` variant, a 10–18% tint for badges, banners and pills. Tinted pills sit on an opaque surface so their text keeps its contrast.

**Brand vs gain.** Lens Green and gain green are the same hue family on purpose: the brand is about growth. To avoid confusion, primary buttons never sit inside price data, and gain text always carries + and ▲.

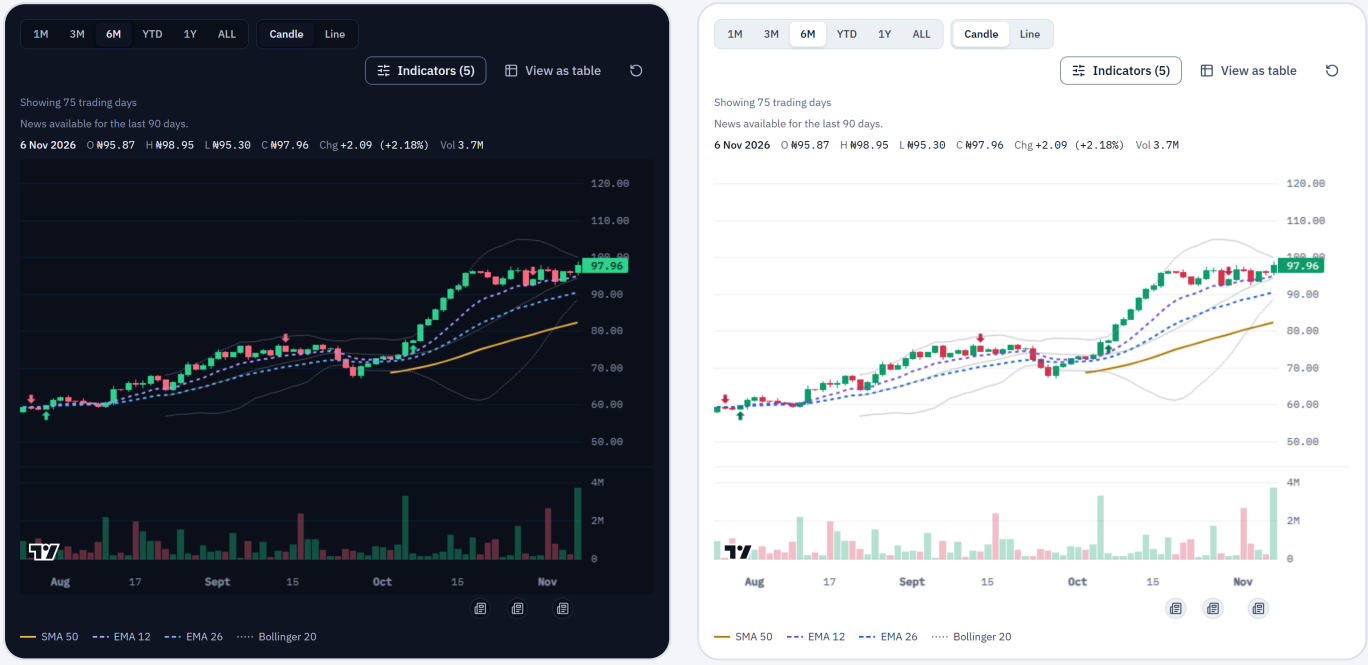
**Not every rise is good.** Stat tiles show their deltas in neutral text. More value traded isn't "good" the way a price gain is, so only prices use gain and loss colours.




Glow belongs to the navigation: the active item's halo in dark mode, and the small market-open dot.

# The price chart

Charts have their own tokens so price, indicators and signals stay readable together. Overlay lines differ by dash pattern as well as colour, so they can be told apart without it.



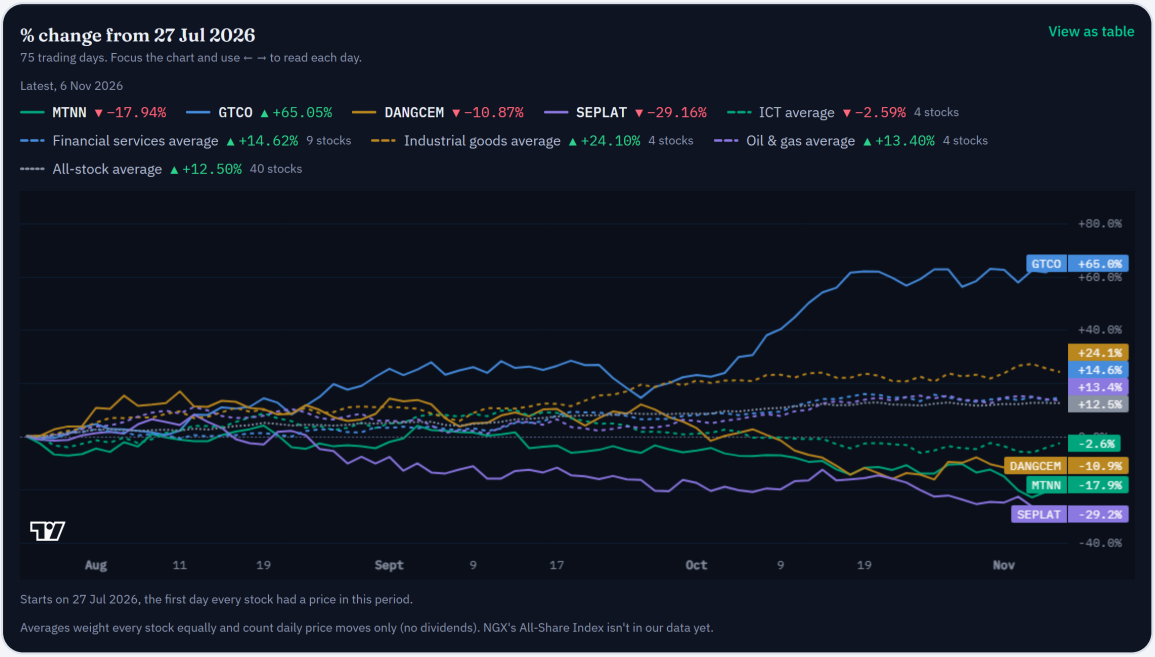

The real chart card in both themes. Markers sit in the price pane. News days appear as newspaper icons in their own band under the time axis, so they never cover a signal. The small logo is the attribution Lightweight Charts requires.

| SERIES | DARK | LIGHT | STYLE |
|---|---|---|---|
| **Up / down candle**<br>`chart-up / chart-down` | #2FD38F<br>#FF6B81 | #0E9F6A<br>#D9334F | Filled bodies |
| **Line (close)**<br>`chart-line` | #00C896 | #009E78 | Solid |
| **SMA 50 / SMA 200**<br>`chart-sma-50 / -200` | #F5B841<br>#FF9F1C | #B07A00<br>#C2410C | Solid |
| **EMA 12 / EMA 26**<br>`chart-ema-12 / -26` | #B49CFF<br>#6CB2FF | #6D4ED8<br>#1D63C9 | Dashed |
| **Bollinger bands**<br>`chart-bollinger` | #8A91A2 | #848C9C | Dotted |
| **Bullish / bearish markers**<br>`chart-signal-buy / -sell` | #2FD38F<br>#FF6B81 | #047A4E<br>#C42343 | Arrow below the bar when bullish, above it when bearish |

The signal tokens keep their `-buy` and `-sell` names, and the API still says `buy` and `sell`. People only ever read “Bullish” and “Bearish”.

# Compare and sector heat

Compare draws up to four stocks and their averages on one chart, so its colours were chosen to stay apart for colour-blind readers.

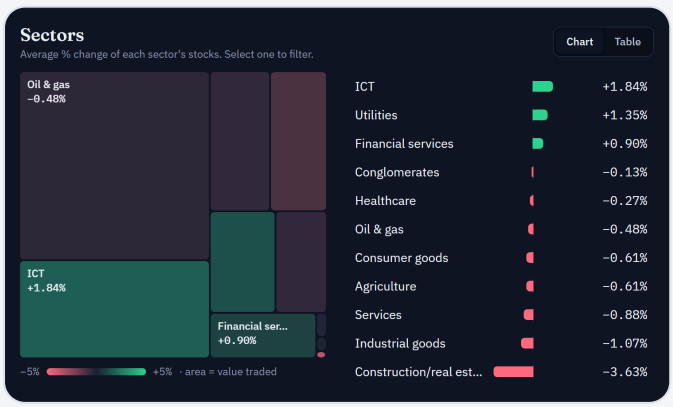



| LINE | LOOKS LIKE | RULE |
|---|---|---|
| **Stocks** Fixed order, never cycled | ——— ——— ——— ——— | Solid, in series 1 to 4. A stock keeps its colour while others are added or removed after it. |
| **Sector average** Optional | - - - - - - - | Dashed, in the colour of the first stock from that sector. |
| **All-stock average** Optional | ..... | Dotted, in the chart axis grey. |

DARK #00A87E #4A8EDE #BC8820 #8E78E2 LIGHT #009E78 #1D63C9 #B07A00 #6D4ED8

Neighbouring colours differ by at least ΔE 15 for deuteranopia, and each has at least 3:1 contrast on the chart background. A legend names every line, so colour is never the only cue.

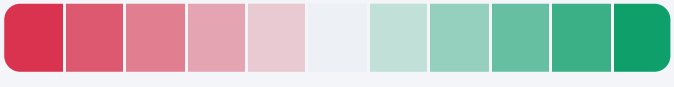
## Sector heat




DARK




LIGHT




Area is value traded and colour is average change, clamped at ±5%: a sector down 8% looks like one down 5%. Ranked bars and a table carry the same numbers.

# Three typefaces, three jobs

IBM Plex Sans carries everything you read, IBM Plex Mono carries every number, and the Fraunces serif gives each page an editorial headline. All three are open source and self-hosted.

**Aa**

**Fraunces 600**

Headlines and titles only. Never body copy, labels, numbers or tickers.

**Aa**

**IBM Plex Sans**

Body, labels, buttons and captions. Weights 400, 500 and 600.

**01**

**IBM Plex Mono**

Numbers, tickers, prices and percentages, with tabular figures. Weights 400 and 500.

## Type roles

| | |
|---|---|
| **Market headline**<br>Fraunces · 36–48 px · 600<br>the landing hero goes to 60 px | **A narrow day for the bulls.** |
| **Page, section and card titles**<br>Fraunces · 30–36 px page,<br>18–24 px section and card | **Stocks**<br>**Where the money went** |
| **Price headline**<br>Plex Mono · 36–48 px · 500<br>tabular figures | ₦97.96 |
| **Body**<br>Plex Sans · 16/24 px · 400 | More stocks rose than fell, 19 to 18, on ₦8.4B traded. |
| **Secondary and table cells**<br>Plex Sans · 14/20 px · 400<br>numbers in cells use Mono | Supporting text and table cells. Price ₦478.50 |
| **Overline and label**<br>Plex Sans · 12/16 px · 500<br>uppercase, short labels only | VALUE TRADED |
| **Numbers and tickers**<br>Plex Mono · inherits its size<br>right-aligned in tables | ₦8.4B · +0.84% · ZENITHBANK |
| **Caption, timestamp, axis**<br>Plex Sans · 12/16 px · 400<br>muted text colour | Data as of 14:35 WAT · 26 Oct |

Sizes are on-screen pixels. Where a range is given, the smaller size is for phones. Samples are shown at the smaller size or close to it.

# Numbers and type rules

A market app is mostly numbers, so they get the most care: one format per kind of value, monospaced tabular figures so columns line up, and a true minus sign.

## Number formats

| KIND | FORMAT | EXAMPLE |
|---|---|---|
| Price | ₦ and 2 decimal places, grouped | ₦1,234.50 |
| Change | arrow, sign, 2 decimals; percent after a dot | ▲ +12.40 · +2.31% |
| Percent | sign, 2 decimals and % | +2.31% · −0.45% · 0.00% |
| Large money | compact, 1 decimal | ₦1.2B · ₦845.3M · ₦12.4K |
| Volume | compact, no currency | 12.4M |
| Date and time | always labelled WAT | 26 Oct 2026 · 14:35 WAT |
| Missing value | em dash | — |

## Why tabular figures

| PROPORTIONAL | TABULAR (PLEX MONO) |
|---|---|
| ₦1,111.11 | ₦1,111.11 |
| ₦8,808.88 | ₦8,808.88 |
| ₦404.40 | ₦404.40 |
| ₦17.10 | ₦17.10 |

Every digit takes the same width, so decimal points line up and a refreshing price does not jitter.

## Minus sign and screen readers

| | |
|---|---|
| −0.45 | -0.45 |
| TRUE MINUS, U+2212 | HYPHEN: TOO SHORT |

Display text uses the true minus sign. Screen-reader text spells values out: “down 3.10 naira, minus 0.45 percent”.

## Putting it together




The stock header: overline in Plex Sans, ticker and price in Plex Mono, company name in Fraunces.

## Rules

- **12 px is the floor** for any text, including chart axes.
- **No letter-spacing** except on uppercase overlines.
- **Self-hosted fonts.** The security policy allows no third-party fonts, and only the weights above ship. The first screen preloads Fraunces 600, Plex Sans 400 and the subset holding ₦, so the headline does not reflow.

# Spacing, layout and grid

Everything sits on a 4 px grid, and the app shell changes shape at two breakpoints. Mobile is designed first.

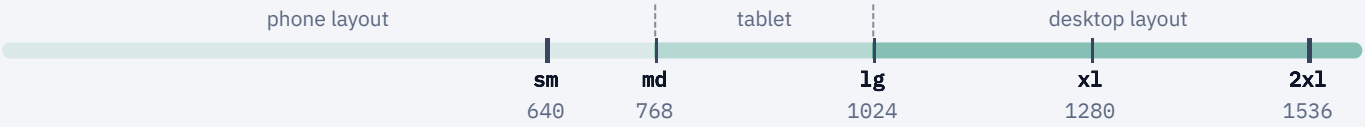
## 4 px spacing scale

| | |
|---|---|
| space-1 | 4 px |
| space-2 | 8 px |
| space-3 | 12 px |
| space-4 | 16 px |
| space-5 | 20 px |
| space-6 | 24 px |
| space-8 | 32 px |
| space-10 | 40 px |
| space-12 | 48 px |
| space-16 | 64 px |
| space-20 | 80 px |

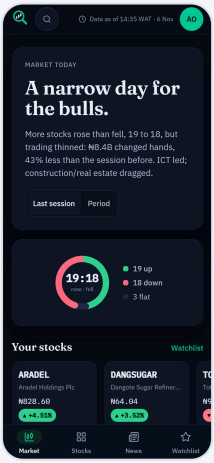
## How spacing is used

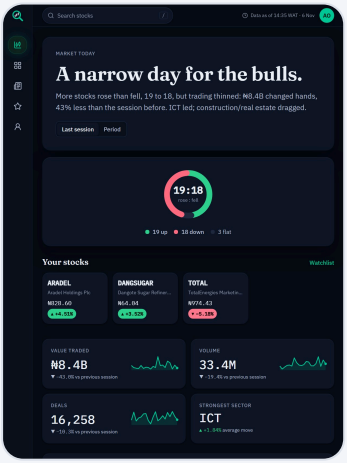
- **Component padding** uses 8, 12 or 16 px.
- **Section gaps** use 24 or 32 px.
- **Gutters** are 16 px on phones, 24 px on tablets and 32 px on desktop.
- **Content** stops growing at 1440 px wide.
- **Grids** use 2, 3 or 4 columns, with fewer on phones.

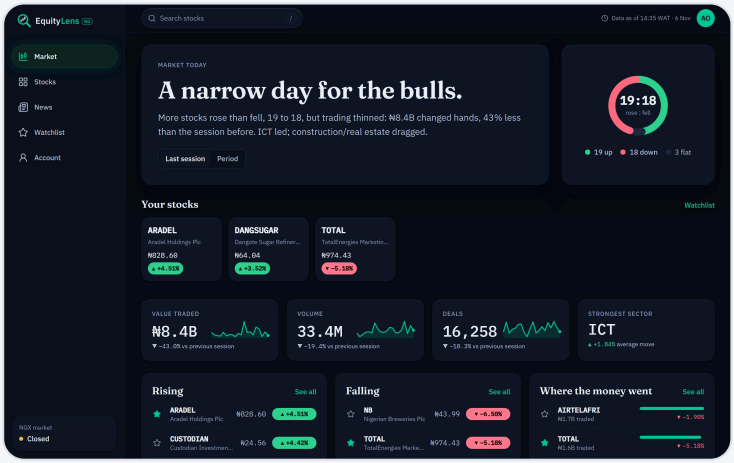
## Breakpoints (Tailwind defaults, px)




## App shell at each size




### Under 768 px

Top bar with the brand mark, search icon, freshness and avatar. Bottom nav: Market · Stocks · News · Watchlist. Account sits in the avatar menu.

### 768 to 1023 px

A 72 px icon rail holds the logo and navigation. The top bar has the full search box, freshness and avatar.

### 1024 px and up

A 240 px sidebar with labels and a market-status card, and a 56 px top bar. Admins also see an Admin item with a waiting count.

# Shape, elevation and motion

Surfaces are separated by depth rather than outlines, corners get rounder as elements get bigger, and motion is quick and quiet.

## Corner radius (px)



**4**
Badges, ticker chips

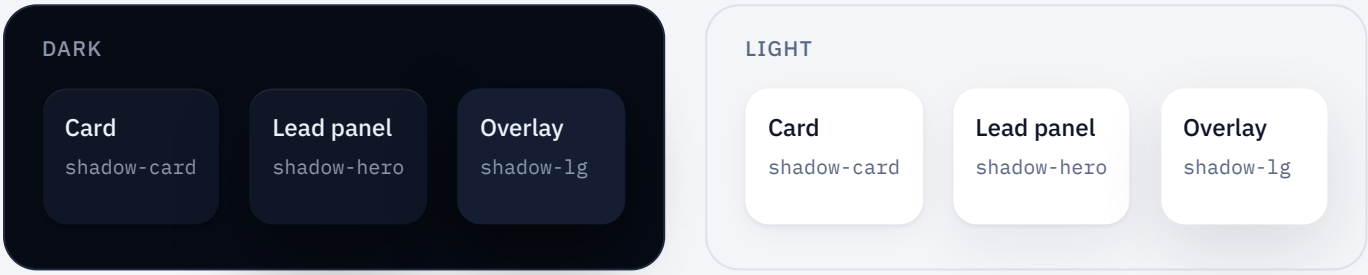
**8**
Buttons, inputs

**16**
Cards, popovers

**20**
Lead panels, dialogs, nav items

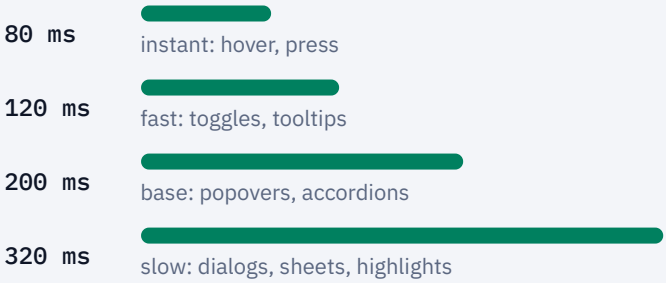
**full**
Pills, avatars

## Elevation




Cards and tiles use `shadow-card`: a hairline top highlight plus a soft shadow in dark mode, a soft shadow in light. The page's lead panel, such as the market headline or stock header, uses `shadow-hero`. Hovering a linked card lifts it from one to the other. Menus and dialogs keep the stronger overlay shadows. Cards keep a transparent border so Windows high-contrast mode can still outline them.

## Motion






**Standard**
most transitions

**Enter**
things arriving

**Exit**
things leaving

LAYERING ORDER (Z-INDEX TOKENS), BOTTOM TO TOP

| |
|---|
| `tooltip 70` |
| `toast 60` |
| `modal 50` |
| `overlay 40` |
| `dropdown 20` |
| `sticky 10` |
| `base 0` |

- **Animate opacity and transform only.** They are cheap to render and never shift the layout.
- **Quiet signals of activity.** A thin brand line pulses at the top of the page while data refreshes.
- **Jump highlight.** Arriving at a news story or glossary term from a link tints it for 2.5 seconds.
- **Reduced motion is honoured.** Durations drop to 0 and pulses stop when the device asks.

# Core controls

These are drawn with the app's own stylesheet and class names, in both themes, so they match the product exactly.

## BUTTONS

VARIANTS

Sign in | Compare | Cancel | Delete

SIZES: 32, 40, 44 PX

Small | Medium | Large

VARIANTS

Sign in | Compare | Cancel | Delete

SIZES: 32, 40, 44 PX

Small | Medium | Large

## BUTTON STATES

Save — Default

Save — Hover

Save — Focus

Save — Active

Save — Disabled

Loading

## FIELDS

Email

you@example.com

We'll send a confirmation link.

Password

•••••

Use at least 10 characters.

Email

you@example.com

We'll send a confirmation link.

Password

•••••

Use at least 10 characters.

## SEGMENTED CONTROLS AND TABS

1M | 3M | 6M | YTD | 1Y | ALL

Candle | Line | Last session | Period

Signals | Peers | Data

1M | 3M | 6M | YTD | 1Y | ALL

Candle | Line | Last session | Period

Signals | Peers | Data

## BANNERS

**Add your email to secure your account**
So you can reset your password if you forget it.

**The data is stale**
The newest prices are older than expected.

**Add your email to secure your account**
So you can reset your password if you forget it.

**The data is stale**
The newest prices are older than expected.

# Data building blocks

Three small pieces appear on nearly every screen: the price change, the freshness stamp and the stat tile. Getting these right does most of the work of making the product feel consistent.

## PriceChange

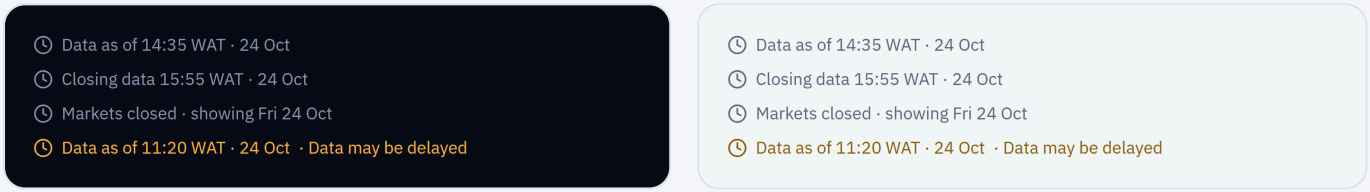



**Inline**
In tables, lists and sentences.

**Pill**
A tinted chip on digest cards.

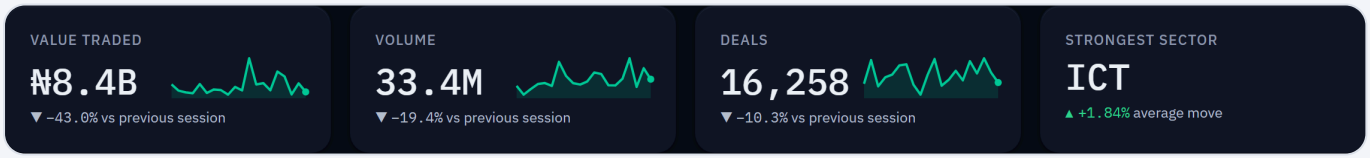
**Solid**
The strongest form, for ranked lists and the stock header. Text on it reaches at least 7:1.

## FreshnessStamp



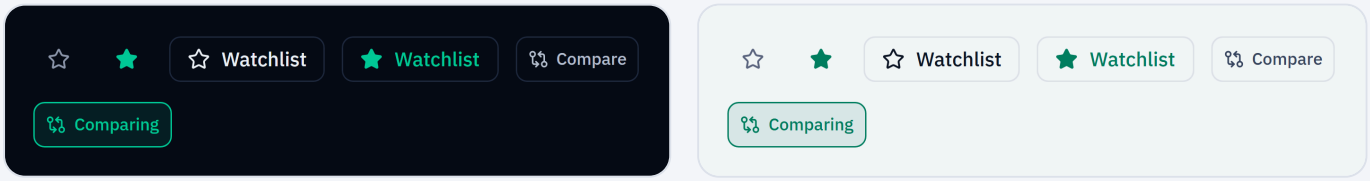

Always in the top bar. From top: during the session, after the 15:55 WAT close, on a weekend or holiday, and when data is more than 2 hours old during market hours.

## Stat tiles




Overline label, Mono value and an optional sparkline. The delta names what it compares against and stays neutral in colour. Sparklines are decorative and hidden from screen readers, because the value is already in the text.

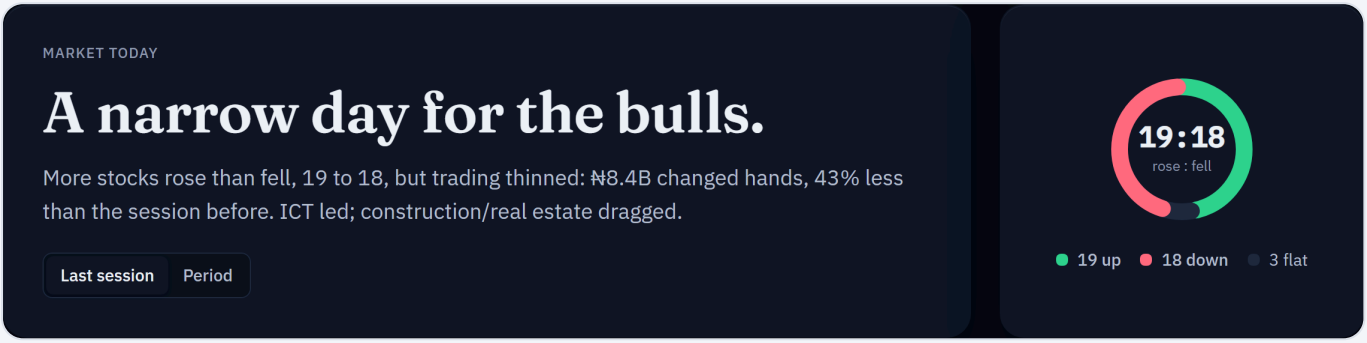
## Stars and Compare toggles



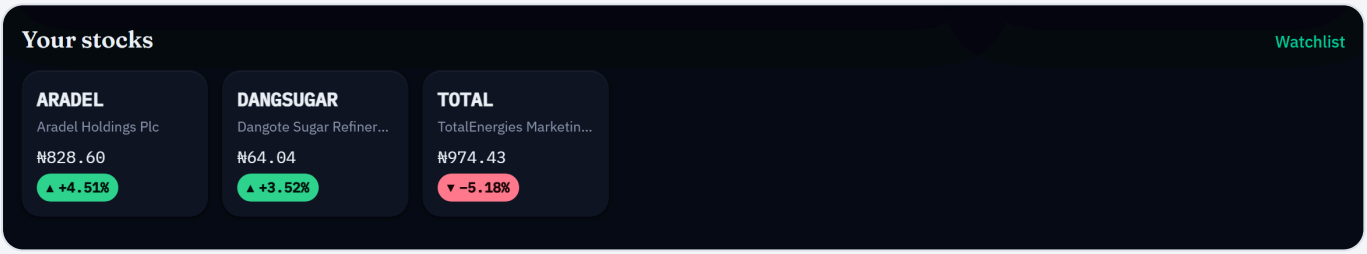

Both act at once. The star puts itself back with a message if the server refuses. The Compare toggle disables itself once four stocks are picked.

# Market Today, piece by piece

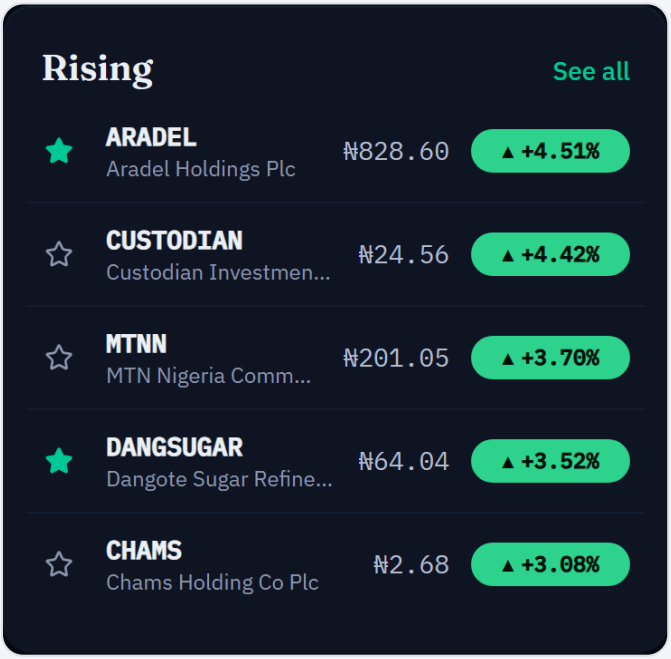
The Market page shows the system at full stretch: an editorial lead, then progressively more detail as you scroll.




**Lead panel and breadth ring.** The headline and standfirst come from a template. The ring splits risers, unchanged and fallers, and its legend spells every number out.




**Your stocks.** Watchlist stocks lead the page as compact cards that scroll sideways.




**Rising and Falling.** Ranked lists use solid pills. A star sits beside each row.

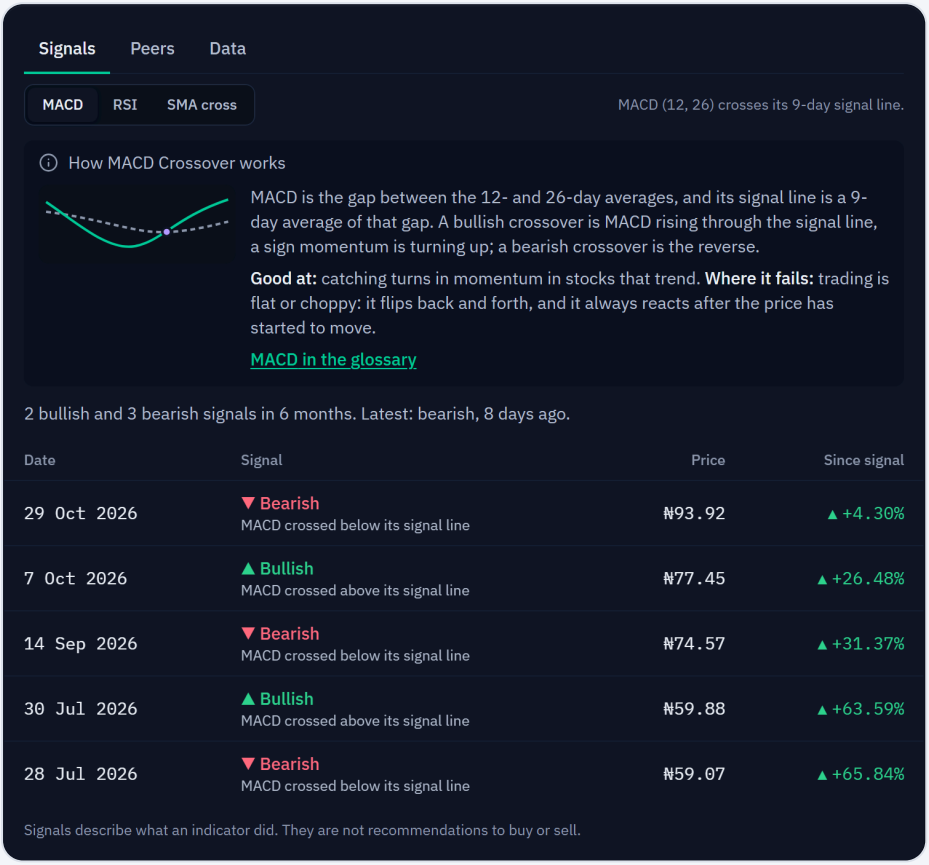



**Where the money went.** A bar shows each stock's share of value traded. On phones the three lists become tabs.

Below these come Sectors (the treemap and ranked bars, shown on page 8), Top stories, and a card leading to every stock on the Stocks page.

# The stock page

One company, with the plain answer first: the price and change, then the chart, then the reasoning behind them. The header is on page 10 and the chart on page 7.



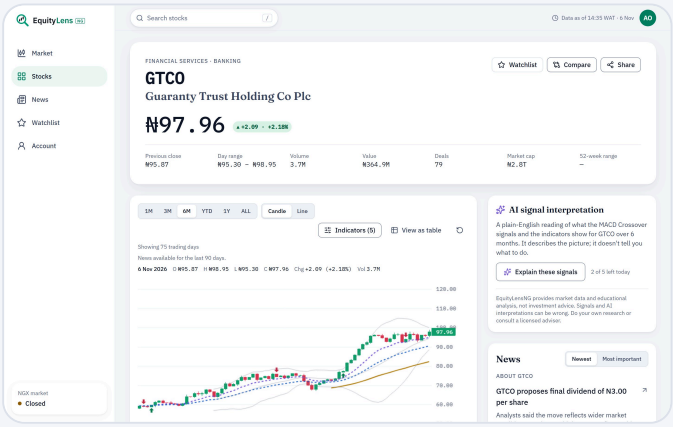

**Signals, Peers and Data tabs.** A strategy switch, a “How MACD Crossover works” explainer with a small diagram, a summary sentence, and the signal list. Selecting a row highlights its marker on the chart.




**Indicators menu.** Every overlay and pane has a one-line explanation, with a News switch at the end.

## Defaults and details

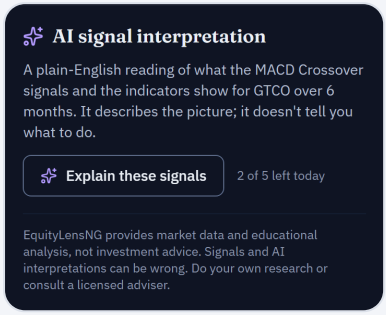
- **Chart.** Candles over 6 months with the volume pane on. Ranges are 1M, 3M, 6M, YTD, 1Y and ALL.
- **Signals.** The list ends with a standing note: “Signals describe what an indicator did. They are not recommendations to buy or sell.”
- **Header actions.** Watchlist, Compare and Share. Share copies the link, or opens the device's share sheet on touch screens.
- **News panel.** “About”, “Mentions” and “Sector news” groups, ordered Newest or Most important.



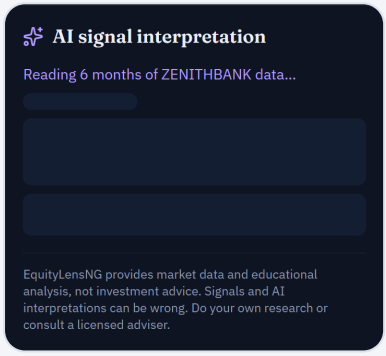

The stock page in the light theme.

# AI, always labelled

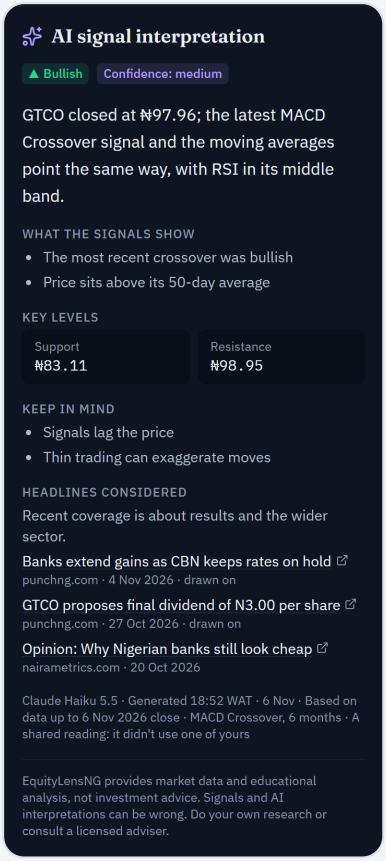
AI output never appears by itself. A person asks for it, it says what it is, and it describes the picture without telling anyone what to do.




**Before.** Says what it will do and how many readings are left today.



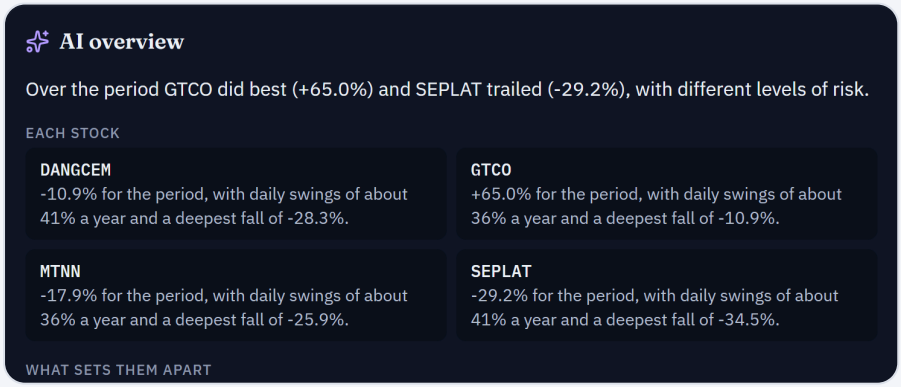

**Loading.** Names what it is reading.




**Reading.** Stance, confidence, summary, drivers, key levels, caveats and headlines. Reading text here is placeholder from the local test provider.

### THE RULES

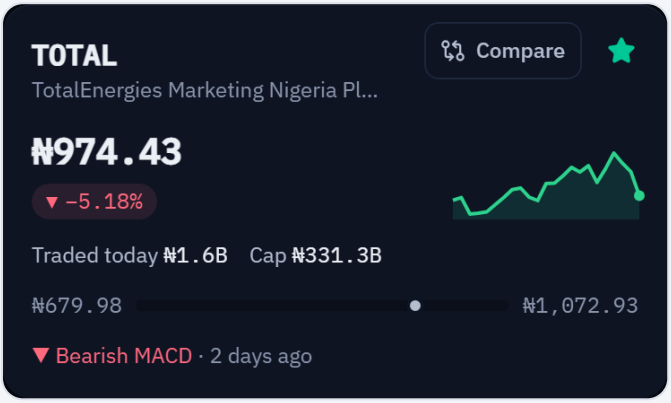
- **Asked for, never automatic.** “Explain these signals” starts it.
- **Always attributed.** Model, time, data date and the disclaimer stay in view, never collapsed.
- **Describes, never advises.** Stance is Bullish, Bearish or Neutral, with no buy or sell language.
- **Checked before showing.** Every reading is validated the same way, whichever provider wrote it.
- **Five readings a day** per person, reset at midnight WAT. A reading someone already asked for is shared and free.
- **Off means hidden.** If AI is switched off, the card disappears and charts carry on.



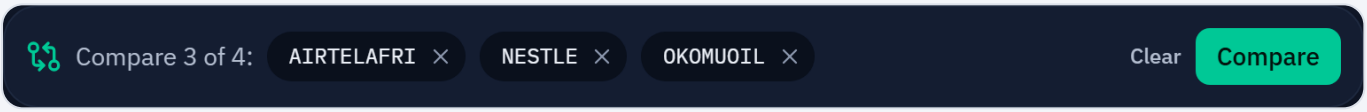

**AI overview on Compare.** The same treatment, and the same daily allowance, for a comparison. It describes how the stocks differ and never which to choose. The default model is Claude Haiku 5.5, and the provider sits behind an interface, so Gemini can be used instead.

# Stocks and Compare

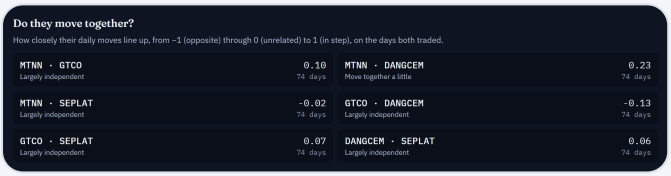
The newest screens, and the clearest example of the system scaling: new pieces, built from the same tokens and rules.




- **Digest card.** Ticker, company and sector, then price with a pill and a sparkline.
- Value traded and market cap, then a bar showing where the price sits in its 52-week range.
- The latest signal in words: “Bearish MACD · 2 days ago”.
- Your watchlist comes first, then the chosen order, most traded by default.

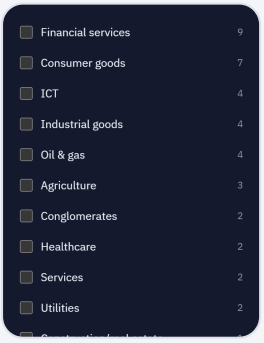



**Compare tray.** Ticking stocks fills a tray fixed to the bottom of the page. One stock compares with its sector.




**Do they move together?** Each correlation also gets plain words, such as “Largely independent”.

- **Filters.** Sector and subsector, size, value traded, latest signal, watchlist only and traded only. Each choice becomes a removable chip.
- **Cards or table.** The table sorts on any column, and its stock column stays pinned when it scrolls sideways on phones.
- **Everything is in the URL,** so any view can be shared.




The Sector filter: a popover with live counts, in the raised surface. Picking a sector narrows it to subsectors.



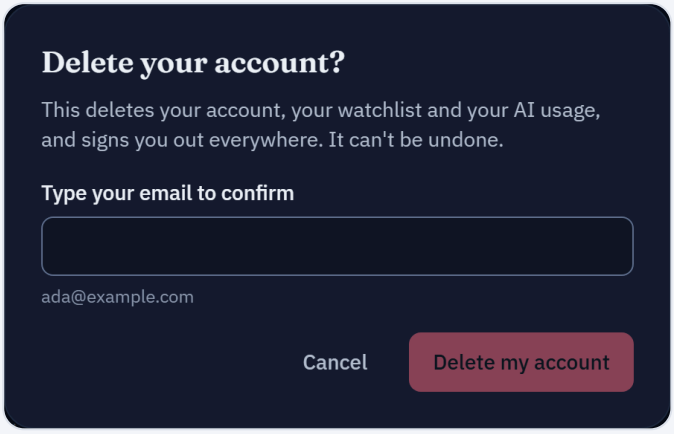

**Side by side.** Returns, risk, trading, size, indicators and signals, grouped under overlines.

# Feedback, overlays and search

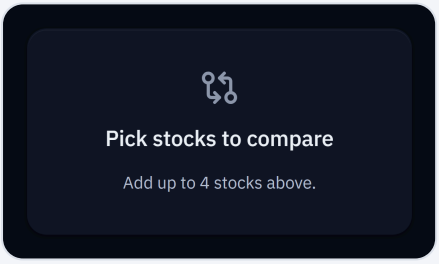
How the product talks back: finding things, confirming actions, and being honest when something is empty or broken.



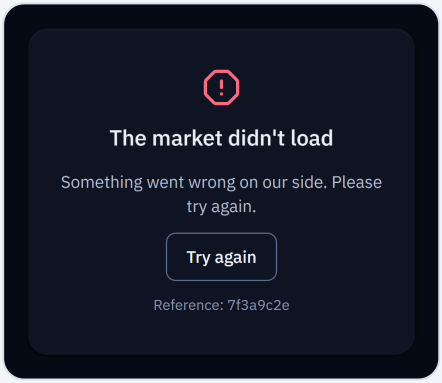

**Search palette.** Press / or Ctrl/⌘ K anywhere. Empty, it offers recent and popular stocks. Full-screen on phones.



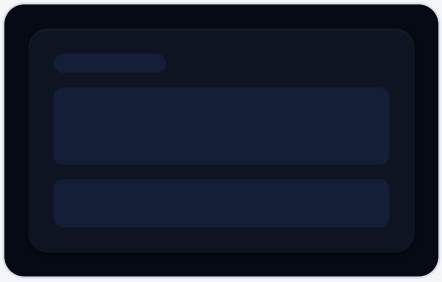

**Destructive actions.** The dialog names the object, and this one asks you to type your email to confirm. Danger sits on the right.



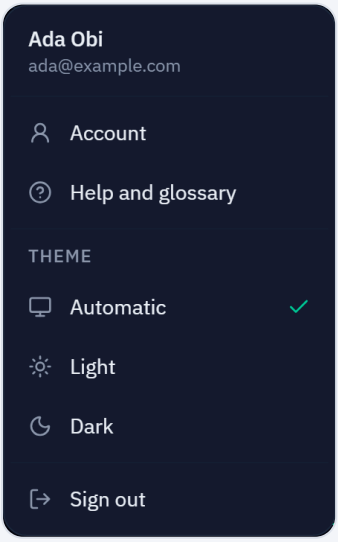

**EmptyState.** Icon, one sentence and one action.



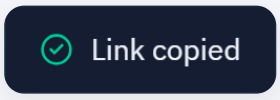

**ErrorState.** Plain cause, retry, and a reference for support.



**Skeleton.** Matches the final layout, so nothing shifts.




**Avatar menu.** Account, Help and glossary, Theme and Sign out.




**Toast.** A brief confirmation that goes after 5 seconds and waits while hovered or focused. It can carry an action such as Undo. It is never used for errors that need action.

**Popovers** hold filters, the Indicators menu and the headlines behind a news marker. They use the raised surface and the 16 px radius.

# Component catalogue

● **Shared** components (25) live in `frontend/src/components/` and serve every screen. ○ **Screen** pieces (26) live with their feature in `frontend/src/features/`. All use tokens only, and Radix supplies accessible behaviour wherever a primitive exists.

## Actions and inputs

| COMPONENT | VARIANTS | KEY BEHAVIOUR |
|---|---|---|
| ● **Button** | primary, secondary, ghost, danger · sm 32, md 40, lg 44 px | A real button; links styled as buttons share its classes. Loading keeps the width, shows a spinner and marks it busy. Never disabled without an explanation nearby. |
| ● **TextField, PasswordField** | default, error · hint, label aside | The label is always visible. Errors are linked to the field. Password fields have a show/hide toggle, and paste and password managers always work. |
| ● **Checkbox, TextArea** | default, error | Labelled controls that handle errors the same way. |
| ● **FormError** | one style | A form-level error, announced as it appears. |
| ● **SymbolSearch** | search button · palette, full-screen on phones | Opens on / or Ctrl/⌘ K. Matches ticker and company name. Arrows move, Enter opens, Esc closes. Recent and popular stocks when empty. |
| ● **SegmentedControl** | md, sm | Single choice from 2 to 6 options, on Radix ToggleGroup. Range, Candle/Line, Last session/Period, cards/table. |
| ● **Tabs** | line | On Radix Tabs, with a brand underline. |
| ● **WatchStar** | icon, labelled | Acts at once and undoes itself with a message if the server refuses. Offers Undo for 5 s on the watchlist page. |
| ○ **PeriodControls** | Last session, Period, presets, custom | Used on Market and Stocks. Custom dates open in a popover with native date inputs. |
| ○ **CompareToggle** | off, on, full | Ticks a stock for Compare and disables at four. |

## Containers and data

| | | |
|---|---|---|
| ● **Card** | one style; lead panels add shadow-elev-2 | Surface with elevation, and a transparent border for forced-colours mode. |
| ● **PageHeader** | title, description, actions | The page title in Fraunces, with an optional supporting line. |
| ● **StatTile** | with delta, with sparkline | Overline, Mono value, neutral delta. Value is 24 px on phones and 30 px above. |
| ● **Sparkline** | any size | A trend line with a flat soft fill. Colour comes from the text colour around it. |
| ● **PriceChange** | inline, pill, solid | Arrow, sign, value and percent. Screen-reader text spells out the direction. |
| ● **FreshnessStamp** | normal, stale | Lives in the top bar. Turns to the warning colour when over 2 hours old in market hours. |
| ● **Badge** | neutral, brand, gain, loss, warning, info, ai, danger | Readable at 12 px. Never the only carrier of meaning. |
| ● **Logo** | lockup, mark | Swaps its dark and light artwork with the theme. |

# Component catalogue, continued

## Feedback and overlays

| COMPONENT | VARIANTS | KEY BEHAVIOUR |
|---|---|---|
| ● **Banner** | info, warning, danger, ai | A left accent bar and icon. Announced as a status, or as an alert when it is danger. |
| ● **Toast** | success, error · with action | Polite announcement. Gone after 5 s, held on hover or focus. |
| ● **Skeleton** | any shape | Pulses, unless reduced motion is on. Matches the final layout. |
| ● **EmptyState** | icon, title, text, action | One sentence and one action. |
| ● **ErrorState** | one style | The cause in plain language, a retry, and a reference for support. |
| ○ **Dialogs** | Radix Dialog, AlertDialog | Used directly per screen. Trap focus, close on Esc, return focus. Bottom sheet for filters on phones. |
| ○ **Popovers** | Radix Popover | Filters, Indicators, custom dates and news markers. |

## Market

| | | |
|---|---|---|
| ○ **BreadthRing** | one style | Risers, unchanged and fallers, with every count in text. |
| ○ **Movers** | columns on desktop, tabs on phones | Rising, Falling and Where the money went, each with “See all”. |
| ○ **SectorTreemap** | Chart, Table | Drawn with d3-hierarchy. Every cell is a focusable button with the full reading in its label. |
| ○ **TopStories** | lead, list | The lead story uses the display serif on the selected surface. |
| ○ **YourStocks** | one style | Watchlist cards that scroll sideways. |

## Stock page

| | | |
|---|---|---|
| ○ **StockHeader** | one style | Ticker, company, price, solid change, stats, and Watchlist, Compare and Share. |
| ○ **PriceChart, ChartCard** | candle, line · Volume, RSI, MACD panes | Lightweight Charts, themed from tokens. ← and → read each day, Home and End jump. “View as table” shows the same data. A spoken summary describes the chart. |
| ○ **NewsStrip** | single, count | Newspaper markers in their own band, with nearby days merged. Hover, focus or tap lists the headlines. |
| ○ **SignalsPanel** | MACD, RSI, SMA cross | Strategy switch, summary sentence and signal list. |
| ○ **StrategyExplainer** | per strategy | What it watches, a small diagram, where it fails, and a glossary link. |
| ○ **AiInsightCard** | idle, loading, reading, out of readings, error | See page 17. |
| ○ **NewsPanel, PeersPanel, DataPanel** | tabs or panels | Grouped news, similar stocks, and the raw numbers. |

# Component catalogue, continued

## Stocks and Compare

| COMPONENT | VARIANTS | KEY BEHAVIOUR |
|---|---|---|
| ○ **StockCards** | digest card | Price, pill, sparkline, value, market cap, 52-week bar and latest signal. |
| ○ **ScreenTable** | sortable | A plain semantic table. Sort state is announced. Stock column pinned when scrolling sideways. |
| ○ **FilterControls, FilterChips** | popovers, chips | Every choice shows as a removable chip, with “Clear all”. All in the URL. |
| ○ **Compare tray** | 1 to 4 stocks | Fixed at the bottom of Stocks, with removable tickers. |
| ○ **CompareChart** | chart, table | % change from a common start, with a legend that reads the hovered day. ← →, Home, End and Esc on the keyboard. |
| ○ **CompareAiCard** | idle, loading, overview | AI overview in the AI treatment, sharing the daily allowance. |

## News

| | | |
|---|---|---|
| ● **NewsItem** | default, lead | External link marked ↗ and announced as opening a new tab. Source, time in WAT, two-line snippet, topic and ticker chips. Never the full article. “Found 3 Nov” when the date is an estimate. |
| ○ **NewsFilters** | desktop rail, mobile sheet | Facets with counts. Zero-count options stay visible but disabled. The sheet ends with a live result count. |
| ○ **Topic and ticker chips** | inline in NewsItem | Topic chips link to filtered news, ticker chips to the stock. |

## App frame

| | | |
|---|---|---|
| ● **AppShell** | sidebar, rail, bottom nav | Skip link, landmarks, current page marked, and the market-status card on desktop. |
| ● **AuthLayout** | one style | A centred card for sign-in, invites and password pages, with the standard disclaimer. |
| ● **UserMenu** | initials | Account, Help and glossary, Theme and Sign out, on Radix DropdownMenu. |

## Admin

| | | |
|---|---|---|
| ○ **Admin pages** | Requests, Invites, Users, Audit log, Usage, Data health | A nav badge counts what is waiting. It is brand-coloured, and turns red only for suspicious sign-ins. |
| ○ **Audit log** | table, filters, CSV | Time in WAT, actor, action, target, outcome and IP. Filters sync to the URL. |

## Icons

**Lucide** icons, usually 20 px, hidden from screen readers unless they stand alone. No emoji in the interface.

# Interaction patterns

Components are only half the system. These rules decide how screens behave when data loads, goes stale, is filtered, shared or fails.

### URL is state

Symbol, range, strategy, overlays, filters, period and comparison all live in the web address, so back, forward and sharing a link all work.

### Loading

Skeletons on first load. While refetching, keep the old data visible with a thin progress line. Never blank a chart that has data.

### Freshness

A FreshnessStamp is always in the top bar. After the 15:55 WAT close it reads “Closing data”. On weekends and holidays: “Markets closed · showing Fri 24 Oct”.

### First run

After accepting an invite, one screen asks “Which stocks do you follow?”. Up to 10 picks join the watchlist and lead Market, or “Skip for now” goes straight there.

### Progressive disclosure

Candle chart, 6 months, volume pane on. RSI, MACD and overlays are a tap away under “Indicators”, and explainers open on request.

### Signals on the chart

The chosen strategy's markers appear on the price chart, and each is linked to its row in the list below.

### AI on demand

A button shows the allowance, such as “3 of 5 left today”. Shared results appear instantly and say they didn't use one of yours.

### Compare

Pick up to four stocks on Stocks, or start from a stock page or its peers. One stock compares with its sector.

### News and external links

News always shows its source and date. External links carry ↗ and say so to screen readers. Relevance is never overstated: “About MTNN”, “Mentions MTNN”, or “Sector news”.

### Filtering lists

Facet counts reflect the current results, and every filter lives in the URL. An empty result names the most restrictive filter, with a one-tap remove.

### Forms

Validate when a field loses focus, then as the user types. On error, focus the first invalid field.

### Destructive actions

A confirm dialog names the object, as in “Delete your account?”, with the danger button on the right.

### Keyboard

/ or Ctrl/⌘ K opens search from anywhere. On charts, ← and → read each day, and Home and End jump to the ends.

### Errors

Explain in plain English what happened and what to do. Never show raw exceptions. If AI fails, charts and signals still work.

# Accessibility

The baseline is WCAG 2.2 level AA, and it is tested rather than assumed.

### Contrast and focus

Contrast is built into the tokens. Every interactive element shows the same 2 px focus ring with a 2 px gap.

### Target size

At least 24 px everywhere, and at least 44 px on touch screens through one shared rule.

### Charts

Every chart has a text summary and a table alternative. Colour is never the only signal.

### Announcements

Price refreshes are not announced, as that would be too noisy. Errors and “AI reading ready” are announced politely.

### Motion

Reduced motion is honoured. Nothing animates on data by itself.

### Language

Pages are marked as Nigerian English, and ₦ is read aloud as “naira”.

### Structure

A skip link, page landmarks, and the current page marked in the navigation.

### Dialogs

Focus is trapped while open and returned on close.

### Links

External links say they open a new tab.

FOCUS RING

Add to watchlist

Sign in

TARGET SIZE (PX)

24 minimum everywhere

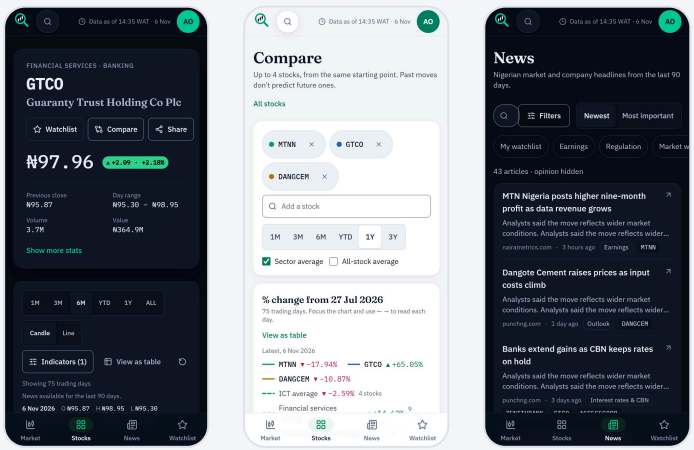
44 minimum on touch screens

## How it is tested

**Automated, now.** Axe runs on every route in both themes in the local Playwright suite, alongside checks for visible focus and reduced motion. It looks for zero serious or critical issues. These tests run before merging and stay out of the repository, so CI itself covers lint, types and builds.

**By hand, before launch.** The core journeys get a manual pass with NVDA on Chrome and VoiceOver on Safari. This is planned for the hardening phase.

## On phones



# Content and voice

The copy follows the same idea as the visuals: plain, calm and specific. It describes what happened, and leaves the decision to the reader.

**DO**

“MTNN rose 2.3% today on ₦4.1B traded.”

**DON'T**

“MTNN is skyrocketing! 🚀”

**DO**

“▲ Bullish MACD · 3 days ago”

**DON'T**

“Buy now.” Never give imperative investment advice.

## Signals are described, never advised

People read “▲ Bullish” and “▼ Bearish” everywhere: on charts, cards, tables, the watchlist and in AI readings. Each signal says what the indicator did, for example “MACD crossed above its signal line”. The interface never tells anyone to buy, sell or hold.

## How the market headline is written

| WHAT HAPPENED | STOCKS UP | STOCKS DOWN |
|---|---|---|
| 70% or more of movers went one way | “A broad rally.” | “Sellers in control.” |
| 55% to 70% went one way | “Buyers had the edge.” | “Sellers had the edge.” |
| Under 55% went one way | “A narrow day for the bulls.” | “A narrow day for the bears.” |
| As many rose as fell | “A split market.” | |
| Nothing moved | “A quiet session.” | |

Rules, not AI. The standfirst is assembled the same way from breadth, value traded against the previous session, and the leading and lagging sectors.

## Explain as you go

Indicators carry a one-line explanation where they are chosen, for example *“RSI above 70 often means a stock has risen quickly and may cool off; below 30, it may be oversold.”* Strategies open a short explainer that says what they are good at and where they fail. Help has a searchable glossary of 42 terms that these link into.

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*EquityLensNG provides market data and educational analysis, not investment advice. Signals and AI interpretations can be wrong. Do your own research or consult a licensed adviser.*

# For developers

How the tokens reach the code, and how the system stays consistent as it changes.

## Tailwind v4 wiring

```
/* frontend/src/styles/globals.css */
@import "tailwindcss";
/* synced by npm run sync:tokens */
@import "./tokens.css";

/* Tokens only: Tailwind's default palette is
   off, so bg-red-500 doesn't exist. */
@theme inline {
  --color-*: initial;
  --color-canvas:  var(--color-bg-canvas);
  --color-surface: var(--color-bg-surface);
  --color-fg:      var(--color-text-primary);
  --color-fg-2:    var(--color-text-secondary);
  --color-line:    var(--color-border-default);
  --color-brand:   var(--color-brand);
  --color-gain:    var(--color-gain);
  --font-headline: var(--font-display);
  --shadow-elev-1: var(--shadow-card);
  --shadow-elev-2: var(--shadow-hero);
  /* …abridged */
}
```

Theming lives in the CSS variables, so utilities such as `bg-surface` switch by themselves. Chart components read tokens through the `useCssVars` hook and re-apply them when the theme changes.

## Helpers to know

| | |
|---|---|
| `.num` | Plex Mono with tabular figures |
| `.eyebrow` | The uppercase overline |
| `.touch-target` | 44 px minimum on touch screens |
| `font-headline` | The Fraunces serif |
| `shadow-elev-1, -2` | Card and lead-panel depth |
| `lib/format.ts` | All number formats |
| `lib/dates.ts` | WAT dates and stamps |

## Governance

1. Tokens change only through a pull request.
2. That PR updates `tokens.css`, the design-system doc and `preview.html` together.
3. It bumps the version in the `tokens.css` header.
4. It adds a `CHANGELOG.md` entry.

A lint rule makes a hex colour in `frontend/src` an error.

## Where things live

| | |
|---|---|
| `design/tokens.css` | All token values. Copied into the frontend at build time. |
| `frontend/src/styles/globals.css` | Token mapping for Tailwind, and the helpers above. |
| `frontend/src/components/` | Shared components: `ui/`, `layout/` and `search/`. |
| `frontend/src/features/` | Screen pieces, one folder per area. |
| `docs/06-design-system.md` | The written spec. |
| `docs/07-brand-guidelines.md` | Logo and brand rules. |
| `python app.py` | Runs the whole app locally on synthetic data, which is how these screens were captured. |

# Status and glossary

This guide describes the built app as of 6 November 2026. Where it differs from the written spec in doc 06, it follows the build.

## In the spec, not built yet

- **Table density.** The “Compact” row toggle.
- **Keyboard extras.** The g m and g w shortcuts, the ? shortcut sheet, and + and – to zoom charts.
- **Price flash.** The optional tint when a number changes.
- **Movers scatter.** Planned for a later release.
- **Tooltip component.** Native titles and popovers stand in for it.
- **Shared table and date picker.** Each screen builds its own for now. Lists are short enough not to need virtualising.

## New in this edition

- **Screens.** Stocks, Compare, Watchlist, Admin, Help, and first-run picks.
- **Components.** AI reading and overview, strategy explainers, news markers, digest cards and the Compare tray.
- **Compare series tokens,** for up to four lines.
- **Wording.** “Bullish” and “Bearish” replace “Buy” and “Sell”.
- **Hours.** NGX now trades 09:30 to 15:55 WAT.
- **AI.** Claude by default, with Gemini optional.
- **Real captures** replace the hand-drawn samples.

## Glossary

**Design token**
A named design value, such as `--color-brand`, used instead of a raw value.

**Radix**
A library of unstyled, accessible React building blocks such as dialogs and tabs.

**WAT**
West Africa Time (UTC+1). NGX trades 09:30 to 15:55 WAT on weekdays.

**SMA, EMA**
Simple and exponential moving averages of price over a number of days.

**Bollinger bands**
A band two standard deviations either side of the 20-day average.

**WCAG 2.2 AA**
The web accessibility standard the app meets. It sets minimum contrast, target sizes and more.

**Tabular figures**
Digits that all take the same width, so columns of numbers line up.

**Breadth**
How many stocks rose compared with how many fell.

**RSI, MACD**
Momentum indicators: Relative Strength Index, and Moving Average Convergence Divergence.

**Standfirst**
The one-sentence summary under a headline, as in a newspaper.